

Package ‘EasyDNB’

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Type Package

Title EasyDNB

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Description Simple use of DNB theory in bulk and single-cell data.

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R topics documented:

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CorandPval

*Calculation of correlations and associated p-values***Description**

A faster, one-step calculation of Student correlation p-values for multiple correlations, properly taking into account the actual number of observations.

Usage

```
CorandPval(
  x,
  y = NULL,
  use = "everything",
  r.na.value = 0,
  p.na.value = 1,
  p.adjust.method = "BH",
  alternative = c("two.sided", "less", "greater"),
  ...
)
```

Arguments

x	a vector or a matrix
y	a vector or a matrix. If NULL, the correlation of columns of x will be calculated.
use	determines handling of missing data. See cor() for details.
p.adjust.method	correction method, a character string. Can be abbreviated. c("holm", "hochberg", "hommel", "bonferroni", "BH", "BY", "fdr", "none")
alternative	specifies the alternative hypothesis and must be (a unique abbreviation of) one of "two.sided", "greater" or "less". the initial letter. "greater" corresponds to positive association, "less" to negative association.
...	other arguments to the function cor().

Details

The function calculates correlations of a matrix or of two matrices and the corresponding Student p-values. The output is not as full-featured as cor.test(), but can work with matrices as input.

Author(s)

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 lzq_calSFNetforCorMatrix

Calculating scale-free network based on correlation matrix

Description

Calculate the adjacency matrix, topological matrix, and identify the optimal soft threshold for defining scale-free network.

Usage

```
lzq_calSFNetforCorMatrix(
  cordata,
  power.vec = c(c(1:10), seq(from = 12, to = 20, by = 2)),
  network.type = "signed",
  RsquaredCut = 0.85
)
```

Arguments

cordata	a correlation matrix.
power.vec	a vector of soft thresholding powers for which the scale free topology fit indices are to be calculated.
network.type	network type. Allowed values are (unique abbreviations of) "unsigned" or "signed".
RsquaredCut	desired minimum scale free topology fitting index R^2 .

Author(s)

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 lzq_cDNB

Conventional DNB analysis

Description

Performing conventional dynamic network biomarker analysis.

Usage

```
lzq_cDNB(
  expr,
  state,
  state.levels,
  cor.method = "pearson",
  p.adjust.method = "BH",
  variation.method = "sd",
  min.size = 10,
  max.size = 2000,
  AddModuleSize = F
)
```

Arguments

<code>expr</code>	A expression dataframe with gene rows and sample columns.
<code>state</code>	A time-series dataframe with two columns, the first is the sample names and the second is the group or time point information.
<code>state.levels</code>	A vector for state sequence.
<code>cor.method</code>	specifies the method for correlation analysis.
<code>p.adjust.method</code>	correction method, a character string. Can be abbreviated. c("holm", "hochberg", "hommel", "bonferroni", "BH", "BY", "fdr", "none")
<code>variation.method</code>	specifies the method for calculating gene variation. sd or cv.
<code>min.size</code>	Minimum gene number of gene modules.
<code>max.size</code>	Maximum gene number of gene modules.
<code>AddModuleSize</code>	Whether to consider gene module size when calculating DNB score.

Author(s)

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 lzq_LcDNB

Landscape conventional DNB analysis

Description

Performing landscape conventional dynamic network biomarker analysis.

Usage

```

lzq_LcDNB(
  expr,
  state,
  state.levels,
  cor.method = "pearson",
  p.adjust.method = "BH",
  variation.method = "sd",
  min.first.neighbor.size = 3,
  min.second.neighbor.size = 1,
  ppi = ppi_h,
  min.combined.score = 900,
  percent = T,
  top.n = 30,
  top.p = 0.05,
  AddModuleSize = F
)
```

Arguments

<code>expr</code>	A expression dataframe with gene rows and sample columns.
<code>state</code>	A time-series dataframe with two columns, the first is the sample names and the second is the group or time point information.
<code>state.levels</code>	A vector for state sequence.
<code>cor.method</code>	specifies the method for correlation analysis.
<code>p.adjust.method</code>	correction method, a character string. Can be abbreviated. <code>c("holm", "hochberg", "hommel", "bonferroni", "BH", "BY", "fdr", "none")</code>
<code>variation.method</code>	specifies the method for calculating gene variation. <code>sd</code> or <code>cv</code> .
<code>min.first.neighbor.size</code>	Minimum size of first order genes of a specific center gene.
<code>min.second.neighbor.size</code>	Minimum size of second order genes of a specific center gene.
<code>ppi</code>	Protein-protein interaction network; background network.
<code>min.combined.score</code>	Minimum combined score for determining protein-protein interaction.
<code>percent</code>	Whether to use Percent to determine the number of DNB genes.
<code>top.n</code>	Only Percent = F takes effect. Center genes with top (number) DNB score were defined as DNB genes.
<code>top.p</code>	Only Percent = T takes effect. Center genes with top (percent) DNB score were defined as DNB genes.
<code>AddModuleSize</code>	Whether to consider gene module size when calculating DNB score.

Author(s)

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 lzq_LDNB

Landscape DNB analysis

Description

Performing landscape dynamic network biomarker analysis.

Usage

```
lzq_LDNB(
  expr,
  state,
  state.levels,
  cor.method = "pearson",
  p.adjust.method = "BH",
  ppi = ppi_h,
  min.combined.score = 990,
  min.first.neighbor.size = 10,
```

```

min.second.neighbor.size = 1,
use.PCC.P.type = "FDR",
use.PCC.P.cutoff = 0.05,
percent = F,
top.n = 30,
top.p = 0.05,
nCores = parallel::detectCores() - 10
)

```

Arguments

<code>expr</code>	A expression dataframe with gene rows and sample columns.
<code>state</code>	A time-series dataframe with two columns, the first is the sample names and the second is the group or time point information.
<code>state.levels</code>	A vector for state sequence.
<code>cor.method</code>	specifies the method for correlation analysis.
<code>p.adjust.method</code>	correction method, a character string. Can be abbreviated. c("holm", "hochberg", "hommel", "bonferroni", "BH", "BY", "fdr", "none")
<code>ppi</code>	Protein-protein interaction network; background network.
<code>min.combined.score</code>	Minimum combined score for determining protein-protein interaction.
<code>min.first.neighbor.size</code>	Minimum size of first order genes of a specific center gene.
<code>min.second.neighbor.size</code>	Minimum size of second order genes of a specific center gene.
<code>use.PCC.P.type</code>	Nominal P value (NP) or adjust P value (FDR) were selected to define the significance of sPCC.
<code>use.PCC.P.cutoff</code>	A cutoff value for use.PCC.P.type.
<code>percent</code>	Whether to use Percent to determine the number of DNB genes.
<code>top.n</code>	Only Percent = F takes effect. Center genes with top (number) DNB score were defined as DNB genes.
<code>top.p</code>	Only Percent = T takes effect. Center genes with top (percent) DNB score were defined as DNB genes.
<code>nCores</code>	The number of cores will be used.
<code>variation.method</code>	specifies the method for calculating gene variation. sd or cv.

Author(s)

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 lzq_SLE

Single-sample landscape entropy analysis

Description

Performing single-sample landscape entropy analysis.

Usage

```
lzq_SLE(
  expr,
  state,
  state.levels,
  cor.method = "pearson",
  p.adjust.method = "BH",
  ppi = ppi_h,
  min.combined.score = 900,
  min.first.neighbor.size = 1,
  percent = T,
  top.n = 30,
  top.p = 0.05,
  nCores = parallel::detectCores() - 10
)
```

Arguments

<code>expr</code>	A expression dataframe with gene rows and sample columns.
<code>state</code>	A time-series dataframe with two columns, the first is the sample names and the second is the group or time point information.
<code>state.levels</code>	A vector for state sequence.
<code>cor.method</code>	specifies the method for correlation analysis.
<code>p.adjust.method</code>	correction method, a character string. Can be abbreviated. c("holm", "hochberg", "hommel", "bonferroni", "BH", "BY", "fdr", "none")
<code>ppi</code>	Protein-protein interaction network; background network.
<code>min.combined.score</code>	Minimum combined score for determining protein-protein interaction.
<code>min.first.neighbor.size</code>	Minimum size of first order genes of a specific center gene.
<code>percent</code>	Whether to use Percent to determine the number of DNB genes.
<code>top.n</code>	Only Percent = F takes effect. Center genes with top (number) DNB score were defined as DNB genes.
<code>top.p</code>	Only Percent = T takes effect. Center genes with top (percent) DNB score were defined as DNB genes.
<code>nCores</code>	The number of cores will be used.

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 lzq_sNMB

Single-sample network module biomarker analysis

Description

Performing single-sample network module biomarker analysis.

Usage

```
lzq_sNMB(
  expr,
  state,
  state.levels,
  cor.method = "pearson",
  p.adjust.method = "BH",
  ppi = ppi_h,
  min.combined.score = 900,
  min.first.neighbor.size = 3,
  percent = T,
  top.n = 30,
  top.p = 0.05,
  nCores = parallel::detectCores() - 10
)
```

Arguments

<code>expr</code>	A expression dataframe with gene rows and sample columns.
<code>state</code>	A time-series dataframe with two columns, the first is the sample names and the second is the group or time point information.
<code>state.levels</code>	A vector for state sequence.
<code>cor.method</code>	specifies the method for correlation analysis.
<code>p.adjust.method</code>	correction method, a character string. Can be abbreviated. c("holm", "hochberg", "hommel", "bonferroni", "BH", "BY", "fdr", "none")
<code>ppi</code>	Protein-protein interaction network; background network.
<code>min.combined.score</code>	Minimum combined score for determining protein-protein interaction.
<code>min.first.neighbor.size</code>	Minimum size of first order genes of a specific center gene.
<code>percent</code>	Whether to use Percent to determine the number of DNB genes.
<code>top.n</code>	Only Percent = F takes effect. Center genes with top (number) DNB score were defined as DNB genes.
<code>top.p</code>	Only Percent = T takes effect. Center genes with top (percent) DNB score were defined as DNB genes.
<code>nCores</code>	The number of cores will be used.

Author(s)

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 lzq_SSPN

Sample-specific perturbation network

Description

Performing sample-specific perturbation network analysis.

Usage

```
lzq_SSPN(
  expr,
  ref.samples,
  cor.method = "pearson",
  p.adjust.method = "BH",
  ppi = ppi_h,
  min.combined.score = 900,
  nCores = parallel::detectCores() - 10
)
```

Arguments

expr	A expression dataframe with gene rows and sample columns.
ref.samples	Samples for constructing the background reference network; Column ID or names.
cor.method	specifies the method for correlation analysis.
p.adjust.method	correction method, a character string. Can be abbreviated. c("holm", "hochberg", "hommel", "bonferroni", "BH", "BY", "fdr", "none")
ppi	Protein-protein interaction network; background network.
min.combined.score	Minimum combined score for determining protein-protein interaction.
nCores	The number of cores will be used.

Author(s)

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 lzq_tDNB

Topological DNB analysis

Description

Performing topological dynamic network biomarker analysis.

Usage

```

lzq_tDNB(
  expr,
  state,
  state.levels,
  cor.method = "pearson",
  p.adjust.method = "BH",
  variation.method = "sd",
  min.size = 10,
  max.size = 2000,
  AddModuleSize = F,
  power.vec = c(c(1:10), seq(from = 12, to = 20, by = 2)),
  network.type = "signed",
  RsquaredCut = 0.85
)

```

Arguments

<code>expr</code>	A expression dataframe with gene rows and sample columns.
<code>state</code>	A time-series dataframe with two columns, the first is the sample names and the second is the group or time point information.
<code>state.levels</code>	A vector for state sequence.
<code>cor.method</code>	specifies the method for correlation analysis.
<code>p.adjust.method</code>	correction method, a character string. Can be abbreviated. <code>c("holm", "hochberg", "hommel", "bonferroni", "BH", "BY", "fdr", "none")</code>
<code>variation.method</code>	specifies the method for calculating gene variation. <code>sd</code> or <code>cv</code> .
<code>min.size</code>	Minimum gene number of gene modules.
<code>max.size</code>	Maximum gene number of gene modules.
<code>AddModuleSize</code>	Whether to consider gene module size when calculating DNB score.
<code>power.vec</code>	a vector of soft thresholding powers for which the scale free topology fit indices are to be calculated.
<code>network.type</code>	network type. Allowed values are (unique abbreviations of) "unsigned" or "signed".
<code>RsquaredCut</code>	desired minimum scale free topology fitting index R^2 .

Author(s)

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 lzq_TSLE

Time-series landscape entropy analysis

Description

Performing time-series landscape entropy analysis.

Usage

```

lzq_TSLE(
  expr,
  state,
  state.levels,
  cor.method = "pearson",
  p.adjust.method = "BH",
  ppi = ppi_h,
  min.combined.score = 900,
  min.first.neighbor.size = 3,
  percent = T,
  top.n = 30,
  top.p = 0.05
)

```

Arguments

<code>expr</code>	A expression dataframe with gene rows and sample columns.
<code>state</code>	A time-series dataframe with two columns, the first is the sample names and the second is the group or time point information.
<code>state.levels</code>	A vector for state sequence.
<code>cor.method</code>	specifies the method for correlation analysis.
<code>p.adjust.method</code>	correction method, a character string. Can be abbreviated. c("holm", "hochberg", "hommel", "bonferroni", "BH", "BY", "fdr", "none")
<code>ppi</code>	Protein-protein interaction network; background network.
<code>min.combined.score</code>	Minimum combined score for determining protein-protein interaction.
<code>min.first.neighbor.size</code>	Minimum size of first order genes of a specific center gene.
<code>percent</code>	Whether to use Percent to determine the number of DNB genes.
<code>top.n</code>	Only Percent = F takes effect. Center genes with top (number) DNB score were defined as DNB genes.
<code>top.p</code>	Only Percent = T takes effect. Center genes with top (percent) DNB score were defined as DNB genes.

Author(s)

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 lzq_TSNMB

Time-series network module biomarker analysis

Description

Performing time-series network module biomarker analysis.

Usage

```

lzq_TSNMB(
  expr,
  state,
  state.levels,
  cor.method = "pearson",
  p.adjust.method = "BH",
  ppi = ppi_h,
  min.combined.score = 900,
  min.first.neighbor.size = 3,
  percent = T,
  top.n = 30,
  top.p = 0.05
)

```

Arguments

<code>expr</code>	A expression dataframe with gene rows and sample columns.
<code>state</code>	A time-series dataframe with two columns, the first is the sample names and the second is the group or time point information.
<code>state.levels</code>	A vector for state sequence.
<code>cor.method</code>	specifies the method for correlation analysis.
<code>p.adjust.method</code>	correction method, a character string. Can be abbreviated. c("holm", "hochberg", "hommel", "bonferroni", "BH", "BY", "fdr", "none")
<code>ppi</code>	Protein-protein interaction network; background network.
<code>min.combined.score</code>	Minimum combined score for determining protein-protein interaction.
<code>min.first.neighbor.size</code>	Minimum size of first order genes of a specific center gene.
<code>percent</code>	Whether to use Percent to determine the number of DNB genes.
<code>top.n</code>	Only Percent = F takes effect. Center genes with top (number) DNB score were defined as DNB genes.
<code>top.p</code>	Only Percent = T takes effect. Center genes with top (percent) DNB score were defined as DNB genes.

Author(s)

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